

- **Install Node.js** Make sure Node.js is installed on your system. You can check with:

```
bash
node -v
npm -v
```

- **Create project folder** Put all the files (`app.js`, `routes/products.js`, `routes/cart.js`, `routes/checkout.js`) inside a project directory.

- **Initialize npm** In the terminal, run:

```
bash
npm init -y
```

- **Install Express** Install the required dependency:

```
bash
npm install express
```

- **Run the server** Start your app with:

```
bash
node app.js
```

app.js – Main Entry Point

js

```
const express = require('express');
const app = express();
const PORT = 3000;
```

```
// Custom logging middleware
app.use((req, res, next) => {
  const timestamp = new Date().toISOString();
  const clientIP = req.ip;
  console.log(`[${timestamp}] ${req.method} ${req.originalUrl} - IP: ${clientIP}`);
  next();
});
```

```
// Import route files
const productRoutes = require('./routes/products');
const cartRoutes = require('./routes/cart');
const checkoutRoutes = require('./routes/checkout');
```

```
// Use routes
app.use('/products', productRoutes);
app.use('/cart', cartRoutes);
app.use('/checkout', checkoutRoutes);
```

```
// Start server
app.listen(PORT, () => {
  console.log(`🚀 Server running at http://localhost:${PORT}`);
});
```

📁 routes/products.js

js

```
const express = require('express');
const router = express.Router();

router.get('/', (req, res) => {
  res.json({
    message: 'Welcome to the Products section!',
    products: ['Shirt', 'Shoes', 'Hat', 'Bag']
  });
});

module.exports = router;
```

📁 routes/cart.js

js

```
const express = require('express');
const router = express.Router();

router.get('/', (req, res) => {
  res.json({
    message: 'Your shopping cart',
    cartItems: ['Shirt', 'Hat']
  });
});

module.exports = router;
```

📁 routes/checkout.js

js

```
const express = require('express');
const router = express.Router();

router.get('/', (req, res) => {
  res.json({
    message: 'Proceeding to checkout',
    status: 'Ready'
  });
});

module.exports = router;
```

 Now Run:

bash

node app.js

You can access:

- <http://localhost:3000/products>
- <http://localhost:3000/cart>
- <http://localhost:3000/checkout>